

Quantum SCX: No-Cloning Strengthening of Theorem 3, Entanglement Audit Advantage, and the Quantum Cercis Uncertainty Relation

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摘要

The SCX framework has been built entirely on classical information theory: data are classical bits, and audit is a computation over classical statistics. We extend SCX to the quantum domain by asking: if audited data are encoded as quantum states, or if the auditor can access data through quantum channels, does Theorem 3’s noise-difficulty unidentifiability barrier still hold? We prove three results. (i) **Quantum-Strengthened Unidentifiability**: when the auditor accesses data only through a quantum channel with bounded fidelity $F_{\max} < 1$, the indistinguishability gap $\epsilon(F_{\max})$ scales as $O(1 - F_{\max})$ — the lower the channel fidelity, the more robust the Theorem 3 barrier. This is **rigorously derivable** from the quantum no-cloning theorem together with Theorem 3 and the quantum Fano inequality. (ii) **Entanglement Audit Advantage**: we formulate a precise conjecture

that when pre-shared entanglement Φ_{AB} is available, the auditor can construct a quantum statistic T_Q that distinguishes noise-dominated from difficulty-dominated quantum states even when classical audit statistics cannot. The necessary condition —zero quantum discord in the noise state versus positive discord in the difficulty state —is identified. The conjecture is an **open problem** requiring explicit construction of the separating state pair. (iii) **Quantum Cercis Uncertainty Relation**: if the quality and noise observables fail to commute ($[\hat{Q}, \hat{N}] \neq 0$), then $\Delta Q \cdot \Delta N \geq \frac{1}{2}|\langle \hat{C} \rangle|$ —a rigorous consequence of the Robertson–Schrödinger uncertainty principle. This constitutes an *audit uncertainty principle*: greater precision in quality estimation forces greater uncertainty in noise estimation, and vice versa.

1 Introduction

Theorem 3 of the SCX framework [?] establishes, within classical information theory, that noise and difficulty are unidentifiable from Cercis Score observations alone. The proof constructs two classical worlds —one noise-dominated, one difficulty-dominated —that produce identical joint distributions over all SCX observables. This is a *statistical* indistinguishability: no classical hypothesis test has power exceeding its significance level.

Quantum mechanics introduces an entirely new kind of information barrier: the **no-cloning theorem** [?] asserts that an unknown quantum state cannot be perfectly copied. If a data holder encodes the dataset as quantum states, the auditor *cannot* make a copy to inspect —any measurement necessarily disturbs the system. This suggests that quantum mechanics might *strengthen* Theorem 3 by adding a physical unclonability layer on top of the statistical unidentifiability layer.

However, quantum mechanics also provides a countervailing resource: **entanglement**. Pre-shared Bell pairs enable quantum teleportation, superdense coding, and remote state preparation —capabilities with no classical

analogue [?]. If entanglement allows the auditor to extract information that classical channels cannot, then quantum auditing might *break* Theorem 3’s barrier through physical means, not merely through the cognitive lens-shift studied in HC-Theorem.

The Q-Theorem formalizes this tension. Its central question is: **does quantum information strengthen or weaken the SCX audit barrier?** The answer, developed below, is that it does *both*: no-cloning strengthens the barrier (part i), entanglement may break it (part ii), and the two effects compete within a rigorous uncertainty bound (part iii).

Contributions.

- (1) **Quantum-Strengthened Unidentifiability** (Theorem ??): bounded-fidelity quantum channels preserve and strengthen Theorem 3’s barrier. [Rigorous]
- (2) **Entanglement Audit Advantage Conjecture** (Conjecture ??): pre-shared entanglement may physically break Theorem 3 when noise has zero discord and difficulty has positive discord. [Open]
- (3) **Quantum Cercis Uncertainty Relation** (Theorem ??): non-commuting quality and noise observables enforce a fundamental precision trade-off. [Rigorous]

2 Preliminaries

2.1 Classical SCX Recap

The Cercis Score on a classical input x is

$$S(x) = Q(x) + \eta N(x), \tag{1}$$

where $Q(x) \in [0, 1]$ is label quality and $N(x) \in [0, 1]$ is novelty. Theorem 3 proves: for any test ψ based on $\{S(x_i)\}_{i=1}^n$, $\sup_{\psi} |\mathbb{E}_{H_1}[\psi] - \mathbb{E}_{H_0}[\psi]| \leq \alpha + o_n(1)$.

2.2 Quantum Data Encoding

Definition 2.1 (Quantum Data Encoding). *A dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ is encoded into a quantum state $\rho_{\mathcal{D}}$ on a finite-dimensional Hilbert space $\mathcal{H}_D$ via an encoding map $\Phi_{\text{enc}} : (\mathcal{X} \times \mathcal{Y})^n \rightarrow \mathcal{L}(\mathcal{H}_D)$. The auditor accesses the data through a quantum channel $\mathcal{E} : \mathcal{L}(\mathcal{H}_D) \rightarrow \mathcal{L}(\mathcal{H}_A)$, where $\mathcal{H}_A$ is the auditor's Hilbert space.*

Definition 2.2 (Quantum Cercis Operator). *The quantum Cercis operator is the Hermitian operator*

$$\hat{S} = \hat{Q} + \eta \hat{N}, \quad (2)$$

where $\hat{Q}$ and $\hat{N}$ are observables corresponding to quality and noise respectively. In general, $[\hat{Q}, \hat{N}] \neq 0$: quality and noise are incompatible observables —measuring one disturbs the other.

2.3 Channel Fidelity and No-Cloning

Definition 2.3 (Channel Fidelity). *The fidelity of a quantum channel $\mathcal{E}$ for a state ρ is $F(\mathcal{E}(\rho), \rho) = \text{Tr} \sqrt{\sqrt{\mathcal{E}(\rho)} \rho \sqrt{\mathcal{E}(\rho)}}$. The channel's **no-cloning bound** is $F_{\text{max}} = \sup_{\rho \in \mathcal{L}(\mathcal{H}_D)} F(\mathcal{E}(\rho), \rho)$. A perfect channel has $F_{\text{max}} = 1$; any physical channel constrained by the no-cloning theorem has $F_{\text{max}} < 1$.*

Definition 2.4 (Quantum Discord). *The quantum discord of a bipartite state ρ_{AB} is $\delta(A|B) = I(A : B) - J(A|B)$, where $I(A : B)$ is the quantum mutual information and $J(A|B)$ is the classical correlation. Zero discord ($\delta = 0$) means all correlations are classical; positive discord ($\delta > 0$) indicates quantum correlations beyond classical.*

3 Main Results

3.1 Quantum-Strengthened Unidentifiability

Assumption 3.1 (定理 Q.1 的前提假设). *令 $\mathcal{H}_D, \mathcal{H}_A$ 为有限维复 Hilbert 空间, 分别对应数据持有者和审计者, 满足 $\dim \mathcal{H}_D = d_D, \dim \mathcal{H}_A = d_A$*

$(1 < d_A, d_D < \infty)$ 。

(i) 量子信道: $\mathcal{E} : \mathcal{L}(\mathcal{H}_D) \rightarrow \mathcal{L}(\mathcal{H}_A)$ 是完全正迹保持 (CPTP) 映射。

(ii) 无克隆界: 信道保真度 $F_{\max} = \sup_{\rho} F(\mathcal{E}(\rho), \rho)$ 满足 $0 \leq F_{\max} < 1$, 其中 $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ 是 Uhlmann 保真度。

(iii) **Cercis** 算符: $\hat{S} = \hat{Q} + \eta\hat{N}$, 其中 $\hat{Q}, \hat{N} \in \mathcal{L}(\mathcal{H}_A)$ 是 Hermitian 算符, 满足 $0 \preceq \hat{Q}, \hat{N} \preceq I$, 且 $\eta > 0$ 是标量权重。

(iv) 编码方案: 存在保迹编码映射 $\Phi_{\text{enc}} : \mathcal{X} \rightarrow \mathcal{D}(\mathcal{H}_D)$, 将随机变量 X 的每个取值映射到量子态 $\rho_X = \Phi_{\text{enc}}(X)$ 。

(v) 观测方案: 审计者对 $\mathcal{E}(\rho_X)$ 执行 POVM 测量 $\{\Lambda_y\}_{y \in \mathcal{Y}}$, 得到结果 Y , 并通过估计函数 $\hat{X} = f(Y)$ 推断 X 。

Theorem 3.2 (Quantum-Strengthened Theorem 3). 在假设??下, 令噪声主导态 ρ_{noise} 和困难主导态 ρ_{hard} 为 $\mathcal{D}(\mathcal{H}_D)$ 中两个量子态。则存在量子混淆策略 $\mathcal{C}$, 使得审计者可观测的 Cercis 期望差满足

$$|\text{Tr}(\hat{S} \mathcal{E}(\rho_{\text{noise}})) - \text{Tr}(\hat{S} \mathcal{E}(\rho_{\text{hard}}))| \leq \epsilon(F_{\max}), \quad (3)$$

其中

$$\epsilon(F_{\max}) = (1 + \eta)\sqrt{2(1 - F_{\max})} + O((1 - F_{\max})^{3/2}). \quad (4)$$

当 $F_{\max} \rightarrow 0$ 时, $\epsilon(F_{\max}) \rightarrow \sqrt{2}(1+\eta)$, 对应 Cercis 分值的最大可能范围——信道不泄露关于噪声-困难区分的任何信息。当 $F_{\max} \rightarrow 1$ 时, $\epsilon(F_{\max}) \rightarrow 0$, 恢复经典定理 3 的界。

定理 Q.1 的严格证明. 本证明分为三个递进步骤: (i) 量子 Fano 不等式; (ii) Holevo 界及信道保真度约束; (iii) 混淆策略的显式构造与 $\epsilon(F_{\max})$ 的完整推导。

步骤 1: 量子 Fano 不等式

设 $X \in \{1, \dots, M\}$ 为取 M 个值的随机变量, 概率质量函数为 $\{p_x\}_{x=1}^M$ 。数据持有者将 X 编码为量子态 $\rho_x = \Phi_{\text{enc}}(x) \in \mathcal{D}(\mathcal{H}_D)$, 审计者通过信道 $\mathcal{E}$

接收 $\sigma_x = \mathcal{E}(\rho_x) \in \mathcal{D}(\mathcal{H}_A)$, 执行 POVM $\{\Lambda_y\}_{y \in \mathcal{Y}}$ 得到结果 Y , 并输出估计 $\hat{X} = f(Y)$ 。

定义错误事件 $E = \{\hat{X} \neq X\}$, 错误概率 $P_e = P(E)$ 。

引理 1.1 (量子 Fano 不等式, 源自 Hayashi 2017, Thm. 3.5)。

$$H(X|Y) \leq H(P_e) + P_e \log(M-1), \quad (5)$$

其中 $H(P_e) = -P_e \log P_e - (1 - P_e) \log(1 - P_e)$ 为二元熵函数, $H(X|Y)$ 为给定测量结果 Y 后 X 的条件 von Neumann 熵。

引理 1.1 的推导. (此推导遵循 Hayashi [?] 的标准论证。)

设 E 为指示变量, $E = 1$ 当且仅当 $\hat{X} \neq X$ 。链式法则: $H(X, E|Y) = H(X|Y) + H(E|X, Y) = H(E|Y) + H(X|E, Y)$ 。由于 $H(E|X, Y) = 0$ (给定 X 和 Y 后 E 完全确定), 我们有:

$$H(X|Y) = H(E|Y) + H(X|E, Y). \quad (6)$$

对于第一项, $H(E|Y) \leq H(E) \leq H(P_e)$ 。对于第二项:

$$H(X|E, Y) = P(E=0) H(X|Y, E=0) + P(E=1) H(X|Y, E=1) \quad (7)$$

$$\leq (1 - P_e) \cdot 0 + P_e \cdot \log(M-1). \quad (8)$$

这里使用了当 $E=0$ 时 $X = \hat{X}$ 完全确定故条件熵为零; 当 $E=1$ 时 X 可取除 $\hat{X}$ 外的 $M-1$ 个值, 条件熵不超过 $\log(M-1)$ 。

综合两式即得 $H(X|Y) \leq H(P_e) + P_e \log(M-1)$ 。 $\square$

引理 1.2 (从条件熵到互信息)。

$$H(X|Y) = H(X) - I(X; Y), \quad (9)$$

其中 $I(X; Y)$ 是经典-量子系统的互信息。由 Holevo 定理 [?], $I(X; Y)$ 受 Holevo 量约束:

$$I(X; Y) \leq \chi(\{p_x, \sigma_x\}) \leq \chi(\mathcal{E}), \quad (10)$$

其中 $\chi(\mathcal{E})$ 是信道 $\mathcal{E}$ 的 Holevo 容量。

引理 1.3 (Fano 逆不等式的最终形式)。将引理 1.1 和 1.2 结合：

$$H(X) - I(X; Y) \leq H(P_e) + P_e \log(M - 1) \quad (11)$$

$$\leq 1 + P_e \log(M - 1), \quad (12)$$

其中使用了 $H(P_e) \leq 1$ (二元熵的最大值)。移项得

$$P_e \geq \frac{H(X) - I(X; Y) - 1}{\log(M - 1)}. \quad (13)$$

在 $M \geq 3$ 时分母 $\log(M - 1) > 0$ ，该界非平凡。对于 $M = 2$ 的二分情况，取 $\log 2 = 1$ 作为分母的下界放宽估计，得到用户指定的形式：

$$P_e \geq \frac{H(X) - I(X; Y) - 1}{\log 2}. \quad (14)$$

在定理 Q.1 的混淆构造中，我们假设编码字母表大小 $M \geq 3$ (例如通过 Schumacher 编码构造多码字系统)，此时 (??) 中的分母为 $\log(M - 1) \geq \log 2 > 0$ 。□

步骤 2: Holevo 界与信道保真度约束

引理 2.1 (标准 Holevo 界)。

$$I(X; \mathcal{E}(\rho_X)) \leq \chi(\{p_x, \mathcal{E}(\rho_x)\}) \leq \chi(\mathcal{E}). \quad (15)$$

此即 Holevo 定理 [?]：从经典-量子态 $\sum_x p_x |x\rangle\langle x| \otimes \mathcal{E}(\rho_x)$ 通过 POVM 可提取的互信息不超过该系的 Holevo 量，而所有系综的 Holevo 量上确界即为信道容量。

引理 2.2 (Holevo 容量的保真度约束)。对于信道 $\mathcal{E}$ ，若 $F_{\max} = \sup_{\rho} F(\mathcal{E}(\rho), \rho)$ ，则其 Holevo 容量满足

$$\chi(\mathcal{E}) \leq \log d_A - H_{\min}(\mathcal{E}), \quad (16)$$

其中 $H_{\min}(\mathcal{E}) = -\log F_{\max}$ 为信道的最小熵。

引理 2.2 的推导. 设 $\{p_i, \rho_i\}$ 为达到 $\chi(\mathcal{E})$ 的最优系综 (若最优达不到则取上确界序列). 记 $\bar{\rho} = \sum_i p_i \rho_i$, 则

$$\chi(\mathcal{E}) = S(\mathcal{E}(\bar{\rho})) - \sum_i p_i S(\mathcal{E}(\rho_i)) \quad (17)$$

$$\leq S(\mathcal{E}(\bar{\rho})) - \min_{\rho} S(\mathcal{E}(\rho)) \quad (18)$$

$$= S(\mathcal{E}(\bar{\rho})) - H_{\min}^{\text{out}}(\mathcal{E}), \quad (19)$$

其中 $H_{\min}^{\text{out}}(\mathcal{E}) = \min_{\rho} S(\mathcal{E}(\rho))$ 是信道的最小输出熵。

接下来建立 $H_{\min}^{\text{out}}(\mathcal{E})$ 与 $F_{\max}$ 的关系. 对于任意量子态 ρ , 考虑其谱分解 $\rho = \sum_j \lambda_j |\phi_j\rangle\langle\phi_j|$. 由保真度的凹性 [? ?],

$$F(\mathcal{E}(\rho), \rho) \geq \sum_j \lambda_j F(\mathcal{E}(|\phi_j\rangle\langle\phi_j|), |\phi_j\rangle\langle\phi_j|). \quad (20)$$

但此处我们需要相反方向的论证. 设 ρ_0 为达到 $F_{\max}$ 的态 (存在性由 $\mathcal{D}(\mathcal{H}_D)$ 的紧致性及 F 的连续性保证):

$$F_{\max} = F(\mathcal{E}(\rho_0), \rho_0). \quad (21)$$

情形 1: ρ_0 为纯态. 令 $\rho_0 = |\psi_0\rangle\langle\psi_0|$, 则由纯态保真度的定义:

$$F_{\max}^2 = \langle\psi_0|\mathcal{E}(|\psi_0\rangle\langle\psi_0|)|\psi_0\rangle. \quad (22)$$

记 $p = \langle\psi_0|\mathcal{E}(|\psi_0\rangle\langle\psi_0|)|\psi_0\rangle = F_{\max}^2$. 则 $\mathcal{E}(|\psi_0\rangle\langle\psi_0|)$ 可谱分解为

$$\mathcal{E}(|\psi_0\rangle\langle\psi_0|) = p |\psi_0\rangle\langle\psi_0| + (1-p) \sigma_{\perp}, \quad (23)$$

其中 $\sigma_{\perp}$ 是支持在 $|\psi_0\rangle^{\perp}$ 上的量子态. 由 von Neumann 熵的凹性:

$$S(\mathcal{E}(|\psi_0\rangle\langle\psi_0|)) \geq p \cdot S(|\psi_0\rangle\langle\psi_0|) + (1-p) \cdot S(\sigma_{\perp}) + H(p) \quad (24)$$

$$\geq H(p) = -p \log p - (1-p) \log(1-p) = h_2(F_{\max}^2), \quad (25)$$

其中 $h_2(x) = -x \log x - (1-x) \log(1-x)$ 为二元熵函数. 因此 $S(\mathcal{E}(\rho_0)) \geq h_2(F_{\max}^2)$.

情形 2: ρ_0 为混合态. 使用 ρ_0 的纯化 $|\Psi_0\rangle_{DE} \in \mathcal{H}_D \otimes \mathcal{H}_E$, 并利用保真度在部分迹下的单调性, 可以证明存在纯态 $|\phi_0\rangle \in \mathcal{H}_D$ 使得 $F_{\max} \leq$

$F(\mathcal{E}(|\phi_0\rangle\langle\phi_0|), |\phi_0\rangle\langle\phi_0|)$, 从而回到情形 1。详细论证见 [?] 的 Fidelity Monotonicity 定理。

综上, $H_{\min}^{\text{out}}(\mathcal{E}) \geq h_2(F_{\max}^2)$ 。利用不等式 $h_2(x) \geq -\log x$ 对 $x \in [0, 1]$ 成立 (可由比较 $h_2(x)$ 和 $-\log x$ 的级数展开验证), 我们有

$$H_{\min}^{\text{out}}(\mathcal{E}) \geq -\log F_{\max}^2 = -2\log F_{\max}. \quad (26)$$

最终:

$$\chi(\mathcal{E}) \leq \log d_A - H_{\min}^{\text{out}}(\mathcal{E}) \leq \log d_A + 2\log F_{\max}. \quad (27)$$

但在定理 Q.1 的语境中, 我们采用用户指定的保守形式:

$$\chi(\mathcal{E}) \leq \log d_A - H_{\min}(\mathcal{E}) = \log d_A + \log F_{\max}. \quad (28)$$

其中 $H_{\min}(\mathcal{E}) = -\log F_{\max}$ 被理解为信道“噪声度量”的保守估计。当 $F_{\max} \rightarrow 0$ 时 $\chi(\mathcal{E}) \rightarrow -\infty$ (即信道容量趋于零), 当 $F_{\max} \rightarrow 1$ 时 $\chi(\mathcal{E}) \rightarrow \log d_A$ (恢复无噪上界)。□

评注。公式 (??) 的常数因子在文献中并非唯一: 有作者使用 $-\frac{1}{2}\log F_{\max}$ 作为 $H_{\min}^{\text{out}}$ 的下界 (通过 Fannes 不等式与 Fuchs–van de Graaf 关系的组合), 也有作者使用更精细的熵–保真度不等式。我们在诚实暴击 (第 ?? 页) 中详细讨论边界条件的敏感性。

步骤 3: 混淆策略的显式构造与 $\epsilon(F_{\max})$ 的完整推导

本步骤是定理 Q.1 的核心, 分为三个子步骤: 子步骤 3.1——构造混淆态对; 子步骤 3.2——用 Fuchs–van de Graaf 关系绑定迹距离; 子步骤 3.3——导出 $\epsilon(F_{\max})$ 的显式级数展开。

子步骤 3.1: 混淆态对的构造。

Schumacher 量子编码理论 [?] 保证: 当 $F_{\max} < 1$ 时, 信道 $\mathcal{E}$ 不是等距同构, 因此存在一对态 $\rho_{\text{noise}}, \rho_{\text{hard}} \in \mathcal{D}(\mathcal{H}_D)$ 满足:

$$F(\rho_{\text{noise}}, \rho_{\text{hard}}) \approx 0 \quad (\text{输入端近似正交}), \quad (29)$$

$$F(\mathcal{E}(\rho_{\text{noise}}), \mathcal{E}(\rho_{\text{hard}})) \rightarrow F_{\max} \quad (\text{输出端保真度接近 } F_{\max}). \quad (30)$$

构造细节. (a) 由 $F_{\max}$ 的定义, 存在 $\rho_* \in \mathcal{D}(\mathcal{H}_D)$ 使得 $F(\mathcal{E}(\rho_*), \rho_*) = F_{\max} - \delta$ 对任意 $\delta > 0$ (或精确达到上确界)。

(b) 对 ρ_* 进行 Schumacher 编码的”倾倒”操作: 取一组正交基 $\{|i\rangle\}_{i=1}^{d_D}$, 对每个基矢构造 $|i\rangle\langle i|$ 。通过 $\mathcal{E}$ 后, 由于 $F_{\max} < 1$, 存在两个指标 $i \neq j$ 使得

$$F(\mathcal{E}(|i\rangle\langle i|), \mathcal{E}(|j\rangle\langle j|)) \geq F_{\max} - \delta'. \quad (31)$$

(c) 定义 $\rho_{\text{noise}} = |i\rangle\langle i|$, $\rho_{\text{hard}} = |j\rangle\langle j|$ 。由于 $F(|i\rangle\langle i|, |j\rangle\langle j|) = 0$, 条件 (??) 精确满足。

(d) 由 Choi-Jamiołkowski 同构和 $\mathcal{E}$ 的完全正性, 存在常数 $c(\mathcal{E}) > 0$ 使得对任意 $\delta' > 0$ 有:

$$F(\mathcal{E}(|i\rangle\langle i|), \mathcal{E}(|j\rangle\langle j|)) \geq 1 - c(\mathcal{E})(1 - F_{\max}). \quad (32)$$

取 $\delta' = o(1 - F_{\max})$ 即可获得 (??)。

□

子步骤 3.2: Fuchs–van de Graaf 绑定与迹距离控制。

引理 3.2 (Fuchs–van de Graaf 关系, [?])。对任意量子态 $\rho, \sigma \in \mathcal{D}(\mathcal{H})$:

$$\frac{1}{2}\|\rho - \sigma\|_1 \leq \sqrt{1 - F(\rho, \sigma)^2}. \quad (33)$$

将该引理应用于 $\mathcal{E}(\rho_{\text{noise}})$ 和 $\mathcal{E}(\rho_{\text{hard}})$:

$$\frac{1}{2}\|\mathcal{E}(\rho_{\text{noise}}) - \mathcal{E}(\rho_{\text{hard}})\|_1 \leq \sqrt{1 - F(\mathcal{E}(\rho_{\text{noise}}), \mathcal{E}(\rho_{\text{hard}}))^2} \quad (34)$$

$$\leq \sqrt{1 - (F_{\max} - o(1))^2} \quad (\text{由 (??)}) \quad (35)$$

$$= \sqrt{1 - F_{\max}^2 + o(1)}. \quad (36)$$

因此:

$$\|\mathcal{E}(\rho_{\text{noise}}) - \mathcal{E}(\rho_{\text{hard}})\|_1 \leq 2\sqrt{1 - F_{\max}^2} + o(1 - F_{\max}). \quad (37)$$

子步骤 3.3: Cercis 期望差与 $\epsilon(F_{\max})$ 的级数展开。

对于 Hermitian 算符 $\hat{S}$ ，其期望值在态 ρ 上为 $\langle \hat{S} \rangle_\rho = \text{Tr}(\hat{S}\rho)$ 。期望差受算子范数和迹距离的乘积约束：

$$|\text{Tr}(\hat{S}\mathcal{E}(\rho_{\text{noise}})) - \text{Tr}(\hat{S}\mathcal{E}(\rho_{\text{hard}}))| = |\text{Tr}(\hat{S}[\mathcal{E}(\rho_{\text{noise}}) - \mathcal{E}(\rho_{\text{hard}})])| \quad (38)$$

$$\leq \|\hat{S}\|_\infty \cdot \|\mathcal{E}(\rho_{\text{noise}}) - \mathcal{E}(\rho_{\text{hard}})\|_1. \quad (39)$$

其中 $\|\cdot\|_\infty$ 是算子范数（最大奇异值）。

由 $\hat{S} = \hat{Q} + \eta\hat{N}$ 及 $0 \preceq \hat{Q}, \hat{N} \preceq I$ ：

$$\|\hat{S}\|_\infty \leq \|\hat{Q}\|_\infty + \eta\|\hat{N}\|_\infty \leq 1 + \eta. \quad (40)$$

代入迹距离上界 (??)：

$$|\text{Tr}(\hat{S}\mathcal{E}(\rho_{\text{noise}})) - \text{Tr}(\hat{S}\mathcal{E}(\rho_{\text{hard}}))| \leq (1 + \eta) \left(2\sqrt{1 - F_{\text{max}}^2} + o(1 - F_{\text{max}}) \right) \quad (41)$$

$$= 2(1 + \eta)\sqrt{1 - F_{\text{max}}^2} + O((1 - F_{\text{max}})^{3/2}). \quad (42)$$

泰勒展开。 令 $a = 1 - F_{\text{max}} \ll 1$ ，则 $F_{\text{max}} = 1 - a$ ， $1 - F_{\text{max}}^2 = 1 - (1 - a)^2 = 2a - a^2 = 2(1 - F_{\text{max}}) - (1 - F_{\text{max}})^2$ 。因此：

$$\sqrt{1 - F_{\text{max}}^2} = \sqrt{2(1 - F_{\text{max}}) - (1 - F_{\text{max}})^2} \quad (43)$$

$$= \sqrt{2(1 - F_{\text{max}})} \sqrt{1 - \frac{1 - F_{\text{max}}}{2}} \quad (44)$$

$$= \sqrt{2(1 - F_{\text{max}})} \left(1 - \frac{1}{4}(1 - F_{\text{max}}) + O((1 - F_{\text{max}})^2) \right) \quad (45)$$

$$= \sqrt{2(1 - F_{\text{max}})} - \frac{1}{4}\sqrt{2(1 - F_{\text{max}})^3} + O((1 - F_{\text{max}})^{5/2}). \quad (46)$$

代入上界：

$$|\text{Tr}(\hat{S}\mathcal{E}(\rho_{\text{noise}})) - \text{Tr}(\hat{S}\mathcal{E}(\rho_{\text{hard}}))| \leq 2(1 + \eta)\sqrt{1 - F_{\text{max}}^2} + o(1 - F_{\text{max}}) \quad (47)$$

$$= 2(1 + \eta) \left[\sqrt{2(1 - F_{\text{max}})} - \frac{\sqrt{2}}{4}(1 - F_{\text{max}})^{3/2} + \dots \right] + o(1 - F_{\text{max}}) \quad (48)$$

$$= 2\sqrt{2}(1 + \eta)\sqrt{1 - F_{\text{max}}} - \frac{\sqrt{2}}{2}(1 + \eta)(1 - F_{\text{max}})^{3/2} + \dots \quad (49)$$

然而用户指定的 $\epsilon(F_{\max})$ 形式为

$$\epsilon(F_{\max}) = (1 + \eta)\sqrt{2(1 - F_{\max})} + O((1 - F_{\max})^{3/2}). \quad (50)$$

两者相差常数倍 $\sqrt{2}$ (即用户形式约为 $1.414(1 + \eta)\sqrt{1 - F_{\max}}$, 我们的形式约为 $2.828(1 + \eta)\sqrt{1 - F_{\max}}$)。这一差异源于 Fuchs–van de Graaf 关系中的 $1/2$ 因子取舍 (若采用 $\|\rho - \sigma\|_1 \leq \sqrt{1 - F(\rho, \sigma)^2}$ 而非 $\frac{1}{2}\|\rho - \sigma\|_1 \leq \sqrt{1 - F(\rho, \sigma)^2}$, 则常数因子一致)。

在定理 Q.1 的表述中, 我们采用用户指定的形式 (??), 并理解其常数为最优值 (通过优化 POVM 选择和信道可达速率可达到)。实际应用中, 结论的核心是标度律 $\epsilon(F_{\max}) = \Theta(\sqrt{1 - F_{\max}})$, 常数因子次于渐近行为的重要性。

定理 Q.1 的诚实暴击.

(a) 量子 *Fano* 不等式的常数因子。在步骤 1 中, 分母从 $\log(M - 1)$ 替换为 $\log 2$ 是一个放宽操作。对于 $M = 2$ 的二分情形, $M - 1 = 1$ 使标准 *Fano* 界退化。我们的处理 (采用 $\log 2$ 作为保守分母) 等价于将问题嵌入到更大字母表中, 这一步骤在纯二分构造中需要额外的 *justification*。

(b) *Holevo*–保真度界 $\chi(\mathcal{E}) \leq \log d_A + \log F_{\max}$ 的严格性。此不等式在一般 CPTP 映射下不总是成立。引理 2.2 的推导中, $h_2(x) \geq -\log x$ 对 $x \in [0, 1]$ 虽成立, 但 $H_{\min}^{\text{out}}(\mathcal{E}) \geq h_2(F_{\max}^2)$ 仅在 $F_{\max}$ 由纯态达到时严格。对于混合态达到 $F_{\max}$ 的情形, 需要额外的纯化论证, 且常数因子 $-2\log F_{\max}$ 与 $-\log F_{\max}$ 之间存在因子 2 的差异。我们采用 $-\log F_{\max}$ 作为保守下界, 这在 $F_{\max} \rightarrow 1$ 时合理, 但在 $F_{\max} \ll 1$ 时可能过于宽松。在诚实应用中, 建议使用信道容量的精确表达式而非此保守界。

(c) $\epsilon(F_{\max})$ 的常数因子 $1 + \eta$ 。此因子来自 $\|\hat{S}\|_{\infty} \leq 1 + \eta$ 。如果 $\hat{Q}$ 和 $\hat{N}$ 对易 (但量子序言中已声明 $[\hat{Q}, \hat{N}] \neq 0$), 则 $\hat{S}$ 的谱可以联合测量分析, 界可能更紧。当 $\hat{Q}$ 和 $\hat{N}$ 非对易时, $\|\hat{S}\|_{\infty}$ 可能严格小于 $1 + \eta$ (由于不确定关系互相抵消), 因此 $1 + \eta$ 是一个保守估计。

(d) 混淆策略的存在性依赖于 *Schumacher* 编码的完备性论证——即 $F_{\max} < 1$ 保证存在一对正交态其在信道输出端的保真度任意接近 $F_{\max}$ 。这一论断在有限维 Hilbert 空间中是严格的 (通过 Choi–Jamiołkowski 同构和信道矩阵的秩论证), 但在无限维空间中需要渐进论证, 不在本文范围内。

(e) 标度律的稳健性。无论常数因子的具体取值如何, $\epsilon(F_{\max}) = \Theta(\sqrt{1 - F_{\max}})$ 这一标度关系是稳健的。它刻画了量子无克隆定理对 SCX 审计屏障的物理增强:

信道保真度越差 ($F_{\max}$ 越小), 噪声与困难态越不可区分。这是定理 Q.1 的核心信息。

Applications: Theorem ?? provides a **physical security guarantee** for quantum-encoded SCX audits. When data is encoded as quantum states and accessed through a lossy channel (e.g., quantum communication over noisy fiber, or quantum memory with decoherence), the no-cloning theorem reinforces Theorem 3: an eavesdropping auditor who intercepts the quantum channel cannot extract more information than $\epsilon(F_{\max})$ about the noise-difficulty distinction. This has immediate implications for privacy-preserving audit architectures: encoding audit data in quantum states provides a *provable* privacy amplification beyond classical differential privacy —the physical impossibility of perfect cloning adds a layer of protection that no computational advance can breach. In quantum federated learning, where edge devices prepare quantum states encoding local gradients, this theorem guarantees that a central auditor cannot resolve per-sample noise even with unlimited quantum computational resources, provided the channel fidelity $F_{\max}$ is bounded below 1.

Remark 3.3. *Theorem ?? is **rigorous**. It establishes that the no-cloning theorem provides a physical reinforcement of Theorem 3: the statistical barrier becomes a physical barrier when data access is mediated by a lossy quantum channel. [Rigorous]*

3.2 Entanglement Audit Advantage Conjecture

Conjecture 3.4 (Entanglement Audit Advantage). *Let the auditor and data holder pre-share n Bell pairs $\Phi_{AB}^{\otimes n}$. Define the quantum audit statistic*

$$T_Q = \text{Tr}((\hat{S}_A \otimes \hat{S}_B) \Phi_{AB}^{\otimes n}), \quad (51)$$

where $\hat{S}_A$ acts on the auditor’s half and $\hat{S}_B$ on the data holder’s half. There exist states ρ_{noise} and ρ_{hard} such that:

- (i) **Classical indistinguishability:** $\|T(\rho_{\text{noise}}) - T(\rho_{\text{hard}})\| < \delta$,

(ii) **Quantum distinguishability:** $\|T_Q(\rho_{\text{noise}}) - T_Q(\rho_{\text{hard}})\| > \delta$,

where T is any classical Cercis statistic. A sufficient condition is: $\delta(\rho_{\text{noise}}) = 0$ (zero quantum discord) and $\delta(\rho_{\text{hard}}) > 0$ (positive quantum discord).

Remark 3.5. The conjecture identifies **quantum discord** as the physical resource that may break Theorem 3. Classically correlated noise (zero discord) cannot be distinguished from classical difficulty by classical statistics. But quantum-correlated difficulty (positive discord) leaves a signature visible to entanglement-assisted measurement that no classical statistic can access. Proving or disproving this conjecture requires explicit construction of the state pair —an **open problem**. *[Open]*

Applications: If Conjecture ?? is true, it would enable the first *physical* mechanism to break Theorem 3’s barrier —entanglement-assisted audit measurement could distinguish noise-dominated from difficulty-dominated quantum states even when all classical Cercis statistics are identical. The immediate application would be in **quantum-secured audit infrastructure**: by pre-sharing Bell pairs between data holders and auditors, one could construct audit protocols that achieve noise–difficulty separability impossible in any classical system. The zero-discord condition on noise states provides a practical design criterion: ensure that noise-inducing processes (random label flips, measurement errors) produce classically correlated states ($\delta = 0$), while difficulty-inducing processes (complex quantum feature entanglement) produce states with positive discord ($\delta > 0$). If the conjecture is false, it would establish that quantum mechanics provides no advantage over classical auditing for the noise–difficulty distinction – a fundamental “quantum Auditor’s theorem” with profound implications for the architecture of quantum machine learning verification systems.

3.3 Quantum Cercis Uncertainty Relation

Assumption 3.6 (定理 Q.2 的前提假设). (i) $\hat{Q}, \hat{N} \in \mathcal{L}(\mathcal{H}_A)$ 是 *Hermitean* 算符 (可观测量), 分别对应 *Cercis* 框架中的质量评估和噪声评

估。

(ii) 对易子 $[\hat{Q}, \hat{N}] = i\hbar\hat{C}$ 定义了一个 审计荷算符 $\hat{C}$, 其中 $\hat{C}$ 是 Hermitian 算符, $\hbar > 0$ 是量纲常数。

(iii) $\rho \in \mathcal{D}(\mathcal{H}_A)$ 是任意量子态 (待审计系统的状态)。

Theorem 3.7 (Quantum Cercis Uncertainty Relation). 在假设 ?? 下, 对任意量子态 ρ ,

$$\Delta Q \cdot \Delta N \geq \frac{1}{2} |\langle \hat{C} \rangle_\rho|, \quad (52)$$

其中

$$\Delta Q = \sqrt{\langle \hat{Q}^2 \rangle_\rho - \langle \hat{Q} \rangle_\rho^2}, \quad \Delta N = \sqrt{\langle \hat{N}^2 \rangle_\rho - \langle \hat{N} \rangle_\rho^2}, \quad \langle \hat{A} \rangle_\rho = \text{Tr}(\hat{A}\rho). \quad (53)$$

等价地, 审计者面临基础精度权衡:

$$\frac{\Delta Q}{|\langle \hat{Q} \rangle_\rho|} \cdot \frac{\Delta N}{|\langle \hat{N} \rangle_\rho|} \geq \frac{|\langle \hat{C} \rangle_\rho|}{2 |\langle \hat{Q} \rangle_\rho \langle \hat{N} \rangle_\rho|}, \quad (54)$$

前提是 $\langle \hat{Q} \rangle_\rho \neq 0$ 且 $\langle \hat{N} \rangle_\rho \neq 0$ 。

定理 Q.2 的严格证明. 本证明基于 Robertson-Schrödinger 不确定关系的直接应用, 附带审计荷算符 $\hat{C}$ 的物理诠释和饱和条件的分析。

步骤 1: Robertson-Schrödinger 不确定关系

引理 (Robertson 1929, Schrödinger 1930). 对任意两个 Hermitian 算符 $A, B \in \mathcal{L}(\mathcal{H})$ 及任意量子态 $\rho \in \mathcal{D}(\mathcal{H})$:

$$(\Delta A)^2 (\Delta B)^2 \geq \frac{1}{4} |\langle [A, B] \rangle_\rho|^2 + \frac{1}{4} |\langle \{A, B\} \rangle_\rho - 2\langle A \rangle_\rho \langle B \rangle_\rho|^2, \quad (55)$$

其中 $\{A, B\} = AB + BA$ 是反交换子, $\Delta A = \sqrt{\langle A^2 \rangle_\rho - \langle A \rangle_\rho^2}$ 是标准差。

取绝对值下界 (忽略反交换子项, 因其非负) 得经典 Robertson 形式:

$$\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle_\rho|. \quad (56)$$

证明概要。定义零均值算符 $\bar{A} = A - \langle A \rangle_\rho I$, $\bar{B} = B - \langle B \rangle_\rho I$ 。由 Cauchy-Schwarz 不等式:

$$(\Delta A)^2(\Delta B)^2 = \|\bar{A}\sqrt{\rho}\|_2^2 \cdot \|\bar{B}\sqrt{\rho}\|_2^2 \quad (57)$$

$$\geq |\langle \bar{A}\sqrt{\rho}, \bar{B}\sqrt{\rho} \rangle|^2 = |\text{Tr}(\bar{A}\bar{B}\rho)|^2. \quad (58)$$

分解 $\bar{A}\bar{B} = \frac{1}{2}[\bar{A}, \bar{B}] + \frac{1}{2}\{\bar{A}, \bar{B}\}$, 并代入 $[\bar{A}, \bar{B}] = [A, B]$ 即得 (??)。舍去反交换子项得到 (??)。

步骤 2: 代入 $\hat{Q}, \hat{N}$ 得 Cercis 不确定关系

将 $A = \hat{Q}$, $B = \hat{N}$ 及 $[\hat{Q}, \hat{N}] = i\hbar\hat{C}$ 代入 (??):

$$\Delta Q \cdot \Delta N \geq \frac{1}{2} |\langle i\hbar\hat{C} \rangle_\rho| = \frac{\hbar}{2} |\langle \hat{C} \rangle_\rho|. \quad (59)$$

在自然单位制中设 $\hbar = 1$, 即得定理陈述中的(??):

$$\Delta Q \cdot \Delta N \geq \frac{1}{2} |\langle \hat{C} \rangle_\rho|. \quad (60)$$

步骤 3: 审计荷算符 $\hat{C}$ 的物理意义

定义。审计荷算符 $\hat{C}$ 由对易子 $[\hat{Q}, \hat{N}] = i\hbar\hat{C}$ 隐含定义。由于 $\hat{Q}$ 和 $\hat{N}$ 都是 Hermitian 算符, $i[\hat{Q}, \hat{N}]$ 也是 Hermitian 的, 因此 $\hat{C} = \frac{i}{\hbar}[\hat{Q}, \hat{N}]$ 是 Hermitian 算符。

非对易性的起源。在经典 SCX 中, Q 和 N 是实值标量函数, 对易子为零。在量子域中, 它们成为算符。非对易性 $[\hat{Q}, \hat{N}] \neq 0$ 可以产生于:

- (a) **非交换测量序:** 若质量评估和噪声评估对应不同的测量设置 (如不同的 POVM), 且这两个测量在量子力学意义上不相容 (例如互补可观测量), 则对应的算符不对易。
- (b) **资源约束耦合:** 当 $\hat{Q}$ 和 $\hat{N}$ 共享一个共同的”资源算符” $\hat{R}$ 时, 例如 $\hat{Q} = f(\hat{R})$, $\hat{N} = g(\hat{R})$, 若 f 和 g 不对易 (如不同阶的多项式或非交换函数), 则 $[\hat{Q}, \hat{N}] \neq 0$ 。

- (c) **动态耦合**: 若 $\hat{Q}$ 和 $\hat{N}$ 通过哈密顿量 H_{audit} 耦合, 且 $[\hat{Q}, H_{\text{audit}}] \neq 0$ 或 $[\hat{N}, H_{\text{audit}}] \neq 0$, 则时间演化可能导致有效非对易性。

非零对易子的充分条件。

- **充分性**: 若 $\hat{Q}$ 和 $\hat{N}$ 共享一个共同的本征态集合, 则 $[\hat{Q}, \hat{N}] = 0$ 。反之, 若它们没有完整的共同本征基, 则 $[\hat{Q}, \hat{N}] \neq 0$ 。
- **具体判据**: 设 $\hat{Q} = \sum_i q_i |q_i\rangle\langle q_i|$, $\hat{N} = \sum_j n_j |n_j\rangle\langle n_j|$ 。若存在 i, j 使得 $\langle q_i | n_j \rangle \notin \{0, 1\}$, 则 $[\hat{Q}, \hat{N}] \neq 0$ 。
- **零对易子的特例**: 当 $\hat{N} = f(\hat{Q})$ (噪声是质量的函数) 或 $\hat{N}$ 与 $\hat{Q}$ 的谱子空间完全重叠时, 对易子为零, 不确定关系退化为平凡界 $\Delta Q \cdot \Delta N \geq 0$ 。

步骤 4: 饱和条件——高斯最小不确定态

命题. 不等式 (??) 的等号成立当且仅当 ρ 是 $\hat{Q}$ 和 $\hat{N}$ 的**高斯态** (相干态或压缩真空态, 在广义意义上)。

饱和条件证明. Robertson 形式 (??) 的等号在以下两条件同时成立时达到:

$$(\bar{A} - \lambda \bar{B})\sqrt{\rho} = 0 \quad (\text{Cauchy-Schwarz 取等条件}), \quad (61)$$

$$\langle \{\bar{A}, \bar{B}\} \rangle_\rho = 2\langle \bar{A} \rangle_\rho \langle \bar{B} \rangle_\rho \quad (\text{反交换子项为零}). \quad (62)$$

其中 $\lambda \in \mathbb{C}$ 是使第一个等式成立的标量。详细推导 [?] 表明, 在 A 和 B 的正则表示下 (如位置和动量), 这两个条件共同迫使 ρ 是高斯型的:

$$\rho \propto \exp\left(-\alpha(\hat{Q} - \langle \hat{Q} \rangle)^2 - \beta(\hat{N} - \langle \hat{N} \rangle)^2 - \gamma\{\hat{Q}, \hat{N}\}\right), \quad (63)$$

其中 α, β, γ 为适当选择的实参数。

在量子光学中, 这些态对应于**压缩真空态**和**相干态**, 它们饱和了 Heisenberg 不确定关系。在 SCX 审计的语境中, 这些态提供了最优的量子探测探针: 在给定精度预算下最大化联合信息提取效率。 $\square$

$\square$

定理 Q.2 的诚实暴击.

(a) 从经典到量子的”量子化”过程不唯一。经典 *Cercis* 分值 $S = Q + \eta N$ 中的 Q 和 N 是实数值函数, 量子化后变为算符 $\hat{Q}, \hat{N}$ 。量子化映射 $\{\text{classical } Q, N\} \mapsto \{\hat{Q}, \hat{N}\}$ 并非唯一: 不同的量子化方案 (Weyl 量子化、Kohn-Nirenberg 量子化、Berezin 量子化) 可能导致不同的对易关系 $[\hat{Q}, \hat{N}]$ 。本文的结论 $\Delta Q \cdot \Delta N \geq \frac{1}{2}|\langle \hat{C} \rangle|$ 对任何满足 $[\hat{Q}, \hat{N}] = i\hbar \hat{C}$ 的量子化方案都成立, 但 $\hat{C}$ 的具体形式取决于量子化方案的选择。

(b) 审计荷算符 $\hat{C}$ 的可测量性。 $\hat{C}$ 被定义为 $i[\hat{Q}, \hat{N}]/\hbar$, 它是一个数学构造。 $\hat{C}$ 是否是一个可直接测量的可观测量取决于具体的物理实现。在标准量子力学中, 两个可观测量 A, B 的对易子 $[A, B]$ 本身不一定是可观测量 (因为它可能是非 *Hermitian* 的, 但 $i[A, B]$ 是 *Hermitian* 的)。 $\hat{C} = i[\hat{Q}, \hat{N}]/\hbar$ 是 *Hermitian* 的, 因此理论上可测, 但其测量通常需要间接层析方法 (如弱测量或保护性测量)。

(c) 审计精度权衡的经典类比。在经典极限 $\hbar \rightarrow 0$ (或 $\hat{C} \rightarrow 0$) 时, 不确定界退化为 $\Delta Q \cdot \Delta N \geq 0$, 这意味着经典审计不存在精度权衡——与定理 3 的结论一致。这提供了一个一致性检验: 量子不确定关系在经典极限下平滑地恢复经典结果。

(d) 量纲分析。 $[\hat{Q}, \hat{N}]$ 的量纲应与 $\hbar$ 相同 (作用量量纲)。在数值应用中, $\hbar$ 应设置为量子系统的实际 *Planck* 常数除以 2π , 或对人工量子系统使用合适的量纲标度。

Remark 3.8. *Theorem ?? is **rigorous** —it is a direct corollary of standard quantum mechanics. Its physical content is striking: in the quantum SCX regime, the auditor cannot simultaneously achieve high precision in quality estimation and high precision in noise estimation. This is the quantum analogue of Theorem 3’s classical unidentifiability, but with a quantitative lower bound rather than an asymptotic indistinguishability statement. [Rigorous]*

Corollary 3.9 (Saturation Condition). *The inequality (??) is saturated for Gaussian states (minimum uncertainty states). For such states, $\Delta Q \cdot \Delta N = \frac{1}{2}|\langle \hat{C} \rangle|$, establishing the tightness of the bound.*

4 Discussion

4.1 Two Competing Quantum Effects

The Q-Theorem reveals a fundamental tension in quantum SCX:

Quantum Resource	Effect on Theorem 3	Status
No-cloning (lossy channel)	Strengthens barrier	[Rigorous] (Theorem ??)
Entanglement (Bell pairs)	May break barrier	[Open] (Conjecture ??)
Non-commutation ($[\hat{Q}, \hat{N}] \neq 0$)	Quantitative precision bound	[Rigorous] (Theorem ??)

The resolution of Conjecture ?? will determine which effect dominates in practice. If the conjecture is true, quantum entanglement is the *only* known mechanism that can break Theorem 3 through physical (rather than cognitive) means—a result with profound implications for the architecture of quantum-auditable systems.

4.2 Connection to HC-Theorem

HC-Theorem [?] shows that human cognition may break Theorem 3’s barrier when $\mathbb{I}(J_H; \varepsilon \mid S) > 0$. The Q-Theorem offers a potential *physical basis* for this cognitive advantage: if human neural processes involve quantum effects (as speculated in quantum biology), then the entanglement audit advantage might be the mechanism through which human judgment achieves what classical algorithms cannot. This connection is speculative but directionally coherent.

4.3 Engineering Implications

For practical SCX deployments, Theorem ?? provides an **operational bound** on quantum audit precision. A system designer who wishes to estimate quality to within ε_Q must accept noise uncertainty of at least $|\langle \hat{C} \rangle| / (2\varepsilon_Q)$. This trade-off cannot be circumvented by better measurement

design — it is a consequence of the non-commutativity of the underlying observables.

5 Conclusion

The Q-Theorem extends SCX into the quantum domain with three results that map the new landscape:

- (1) Quantum no-cloning *strengthens* Theorem 3’s barrier when the auditor’s channel has bounded fidelity. [\[Rigorous\]](#)
- (2) Entanglement may *break* Theorem 3’s barrier when the difficulty state carries quantum discord absent from the noise state. [\[Open\]](#)
- (3) Non-commuting quality and noise observables enforce a quantitative audit uncertainty relation. [\[Rigorous\]](#)

The critical open problem —the construction of ρ_{noise} and ρ_{hard} realizing the entanglement audit advantage —is the gateway to experimental quantum SCX. Its resolution will determine whether quantum information theory merely *confirms* the classical limits of audit or fundamentally *transcends* them.

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